U24AI049

Assignment 7

The 8 queen's problem is stated as: Place 8 queens on a chessboard such that no two queens attack each other. Considering this problem, complete the following tasks considering a valid heuristic function with justification.

Q1 Generate 50 random initial boards. Run steepest -ascent hill climbing on each and record initial heuristic value, final heuristic value, number of steps taken and status solved or fail. Prove the presence of a local minimum.

Board	Initial H	Final H	Steps	Status
1	10	1	4	Fail
2	11	0	5	Solved
3	4	2	2	Fail
4	6	1	2	Fail
5	5	0	4	Solved
6	11	1	3	Fail
7	4	2	1	Fail
8	9	1	3	Fail
9	7	1	4	Fail
10	6	2	2	Fail
11	10	1	4	Fail
12	9	1	3	Fail
13	8	2	2	Fail
14	6	2	3	Fail
15	12	2	3	Fail
16	5	1	2	Fail
17	6	1	3	Fail
18	6	1	3	Fail
19	8	2	4	Fail
20	13	1	5	Fail
21	8	2	2	Fail
22	8	2	3	Fail
23	9	2	3	Fail
24	7	2	3	Fail
25	6	2	2	Fail
26	8	1	3	Fail
27	5	0	4	Solved
28	11	0	4	Solved
29	12	2	4	Fail

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  32    9          1         4       Fail
  33    9          2         3       Fail
  34    9          1         4       Fail
  35    12         1         5       Fail
  36    6          0         3       Solved
  37    6          3         2       Fail
  38    9          0         5       Solved
  39    6          1         3       Fail
  40    10         0         6       Solved
  41    7          1         4       Fail
  42    6          1         4       Fail
  43    10         1         3       Fail
  44    11         0         5       Solved
  45    6          2         3       Fail
  46    12         0         5       Solved
  47    5          1         2       Fail
  48    10         2         4       Fail
  49    8          0         4       Solved
  50    9          3         2       Fail
--------------------------------------------------
Total Solved: 10/50
Total Failed (Local Minima): 40/50

--- Proof of Local Minimum ---
Board: [1, 0, 4, 3, 3, 2, 1, 1]
Heuristic value at this board: 1
Checking all neighbors...
No neighbor has a lower heuristic value.
=> This is a confirmed LOCAL MINIMUM.
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Q2 Repeat the above experiment on first choice, random restart and simulated annealing variants of hill climbing and compare your results.

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=== First-Choice Hill Climbing ===
Board | Initial H | Final H | Steps | Status
-----------------------------------------------
  1     10          1         8       Fail
  2     11          2         6       Fail
  3     4           2         2       Fail
  4     6           1         5       Fail
  5     5           1         4       Fail
  6     11          2         6       Fail
  7     4           2         2       Fail
  8     9           2         4       Fail
  9     7           2         3       Fail
  10    6           1         4       Fail
  11    10          1         6       Fail
  12    9           2         6       Fail
  13    8           2         6       Fail
  14    6           1         4       Fail
  15    12          1         8       Fail
  16    5           1         3       Fail
  17    6           1         4       Fail
  18    6           1         3       Fail
  19    8           2         4       Fail
  20    13          2         8       Fail
  21    8           1         7       Fail
  22    8           1         4       Fail
  23    9           2         7       Fail
  24    7           2         4       Fail
  25    6           2         3       Fail
  26    8           0         7       Solved
  27    5           0         3       Solved
  28    11          2         7       Fail
  29    12          1         6       Fail
  30    13          2         7       Fail
  31    11          2         6       Fail
  32    9           2         4       Fail
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  46    12          0         6       Solved
  47    5           2         2       Fail
  48    10          2         6       Fail
  49    8           1         5       Fail
  50    9           3         4       Fail
Solved: 6/50 | Avg Steps: 5.1

=== Random Restart Hill Climbing ===
Board | Final H | Total Steps | Restarts | Status
-------------------------------------------------------
  1     0         32            11         Solved
  2     0         34            10         Solved
  3     0         17            5          Solved
  4     0         13            4          Solved
  5     0         57            18         Solved
  6     2         59            20         Fail
  7     0         11            4          Solved
  8     0         23            8          Solved
  9     0         7             2          Solved
  10    0         7             2          Solved
  11    0         48            16         Solved
  12    0         21            6          Solved
  13    0         19            6          Solved
  14    0         5             1          Solved
  15    0         14            5          Solved
```

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  43    0          34            11          Solved
  44    0          29            8           Solved
  45    0          5             1           Solved
  46    0          23            7           Solved
  47    0          11            3           Solved
  48    0          18            5           Solved
  49    0          20            6           Solved
  50    0          30            8           Solved
Solved: 46/50 | Avg Steps: 24.1

=== Simulated Annealing ===
Board | Initial H | Final H | Steps | Status
-------------------------------------------------
  1     10          2         157     Fail
  2     11          1         157     Fail
  3     4           1         157     Fail
  4     6           1         157     Fail
  5     5           1         157     Fail
  6     11          1         157     Fail
  7     4           2         157     Fail
  8     9           2         157     Fail
  9     7           1         157     Fail
  10    6           2         157     Fail
  11    10          1         157     Fail
  12    9           2         157     Fail
  13    8           1         157     Fail
```

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  44    11          1          157      Fail
  45    6           2          157      Fail
  46    12          1          157      Fail
  47    5           1          157      Fail
  48    10          1          157      Fail
  49    8           0          56       Solved
  50    9           1          157      Fail
Solved: 2/50 | Avg Steps: 154.0

=== Comparison Summary ===
Algorithm                    Solved    Avg Steps
----------------------------------------------
First-Choice                   6/50         5.1
Random Restart                46/50        24.1
Simulated Annealing            2/50       154.0

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